SLTC REASEARCH UNIVERSITY

SEMESTER 05
Assignment - 01

Bachelor of Applied Information Technology – (BAIT)

CIT308 – Data Visualization

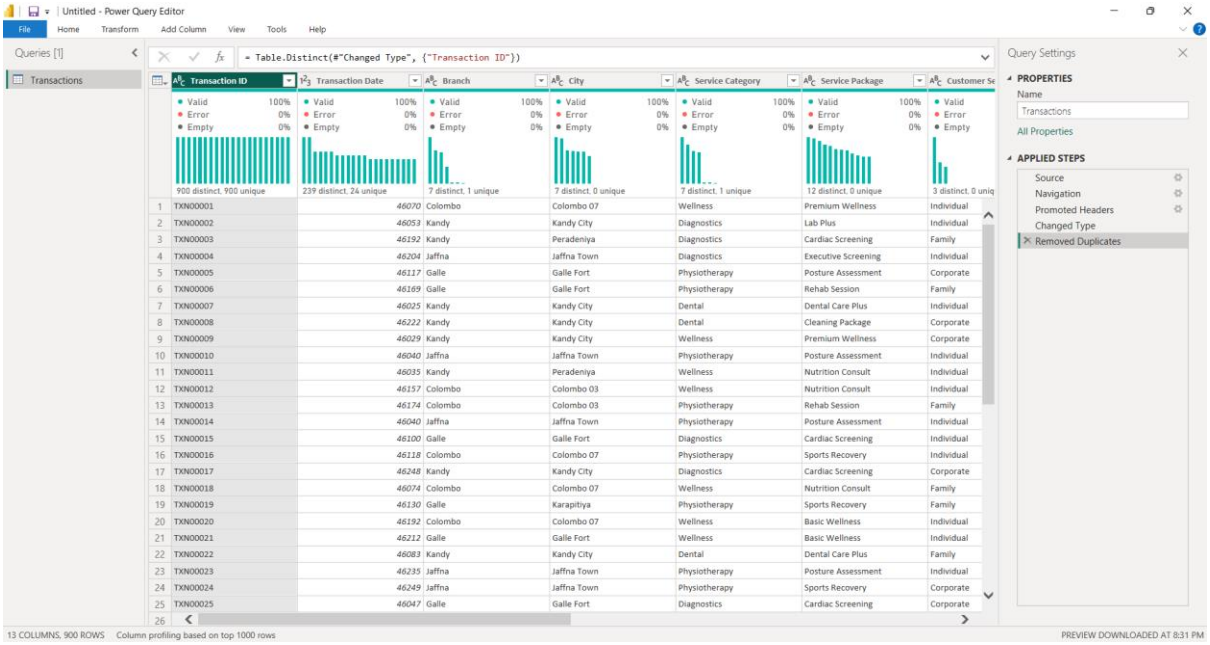
Name: MM Fathima Sumla

Registration No: 23DA2-0852

Data Preparation — Power Query Cleaning Evidence

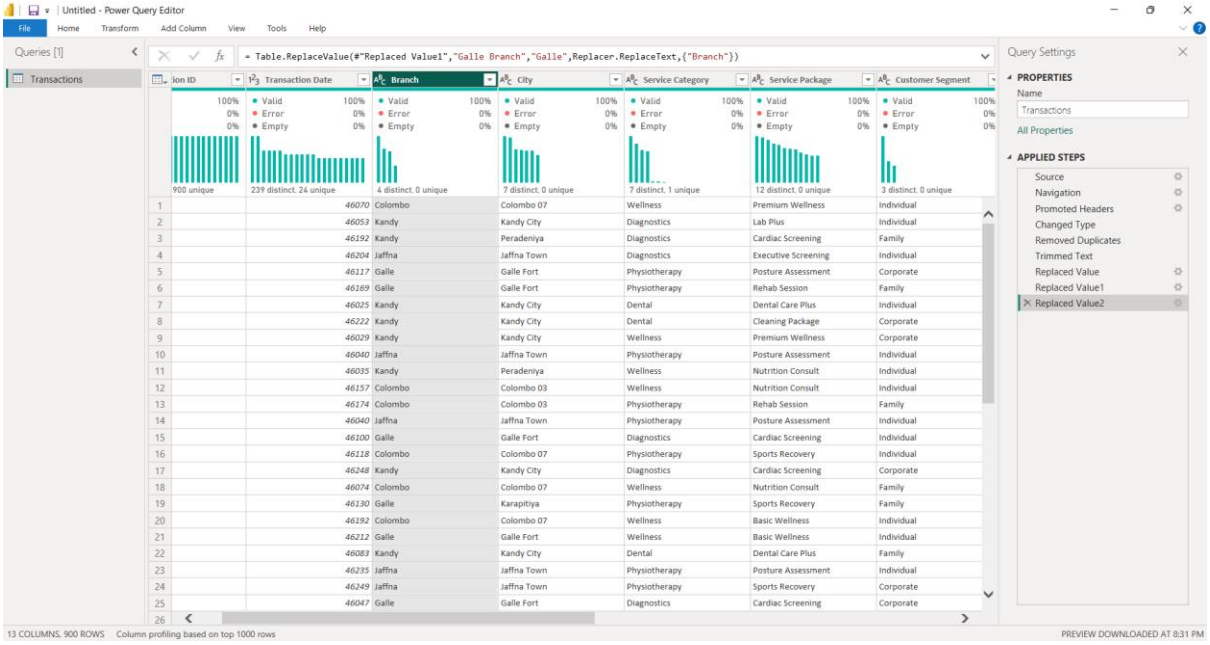
1. Removing duplicate transactions

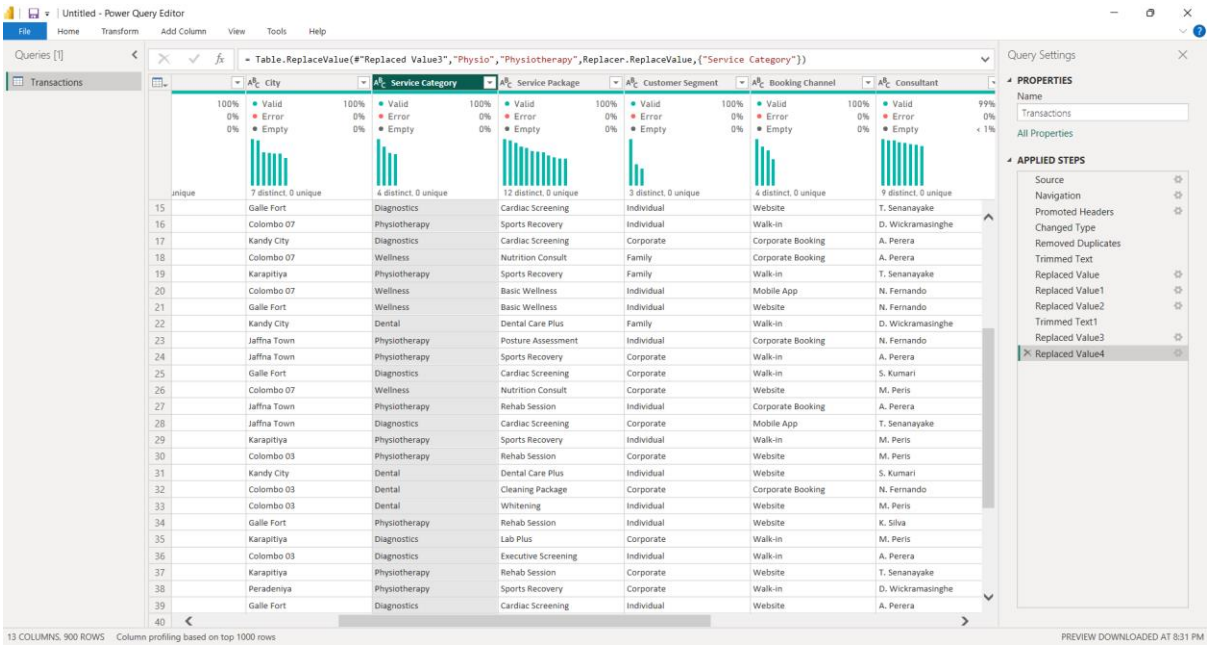
Duplicate Transaction IDs were removed using Table.Distinct on the Transaction ID column, reducing the table from 907 to 900 rows.



2. Standardising Branch and Service Category labels

Inconsistent labels (e.g. "KANDY", "Colombo", "Galle Branch" and "Diagnostic", "Physio") were trimmed and replaced so each field resolves to exactly its correct category.

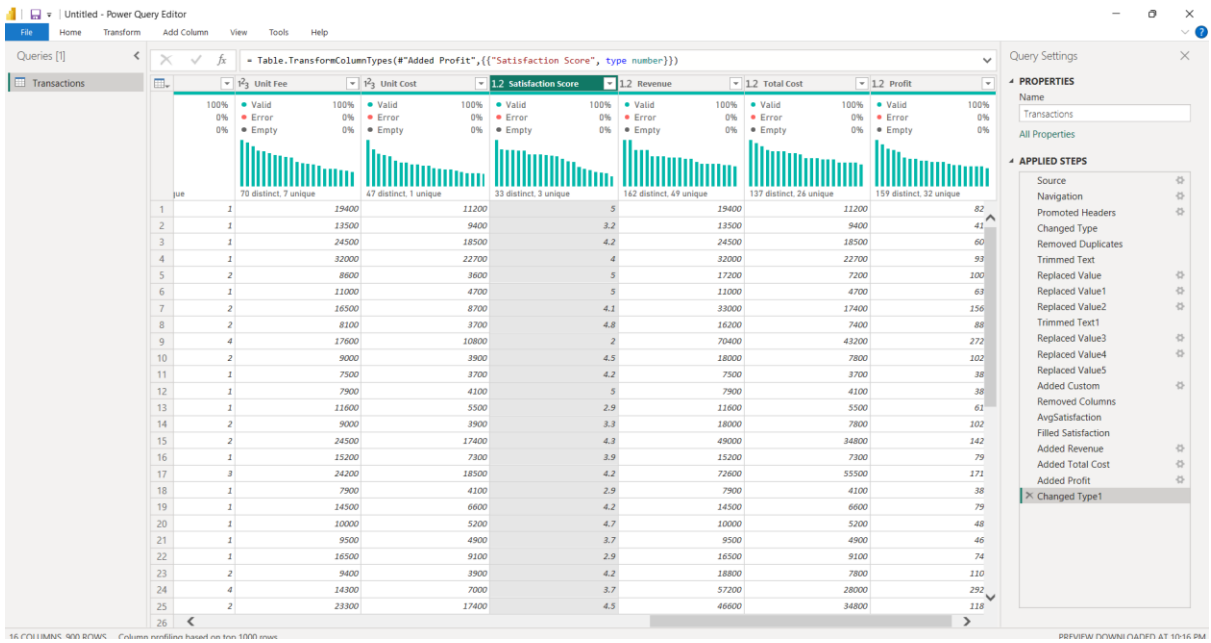




3. Handling missing Consultant and Satisfaction Score values

- Missing Consultant values (8 rows) were replaced with "Unassigned" rather than deleted, to preserve real revenue rows.
- Missing Satisfaction Score values (5 rows) were filled with the column average, since deleting them or using 0 would distort the Average Satisfaction KPI.

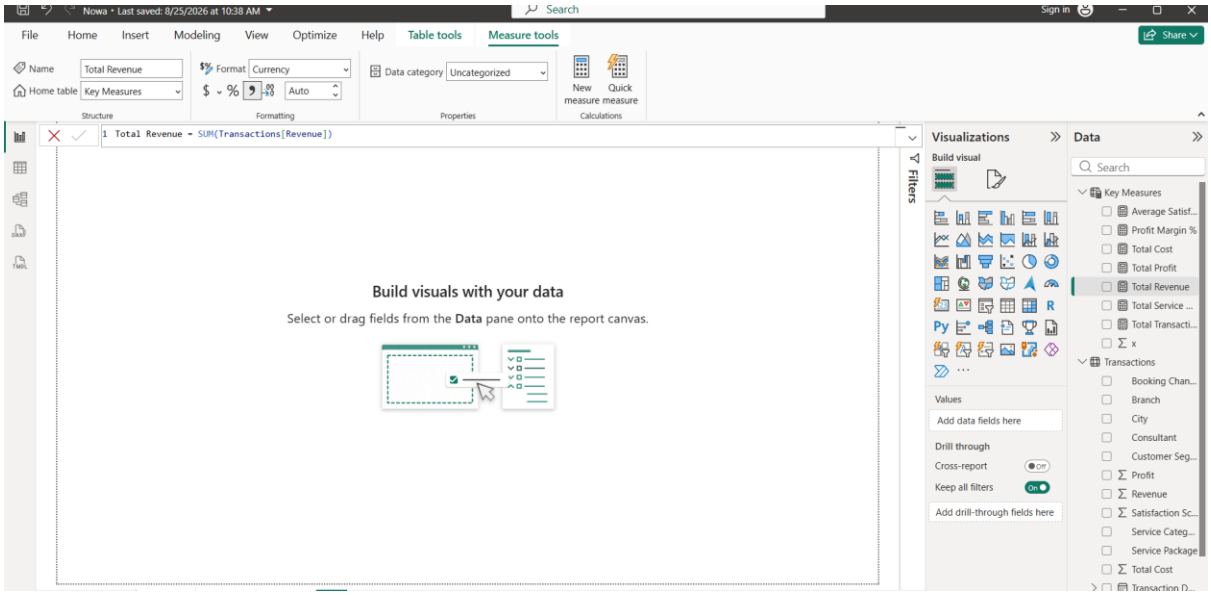
Data types were also corrected — Transaction Date to Date, numeric fields to Whole/Decimal Number — and three calculated columns (Revenue, Total Cost, Profit) were added at the transaction level.



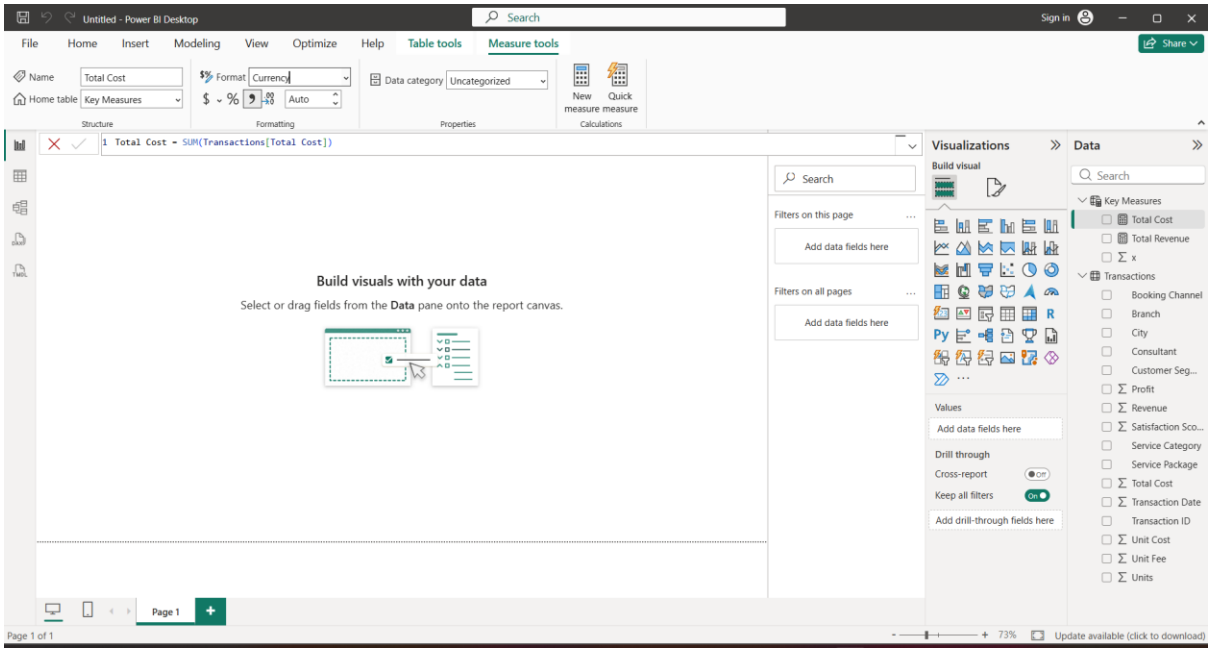
DAX Measures

All seven required measures were created in a dedicated "Key Measures" table so they stay separate from the Transactions data table.

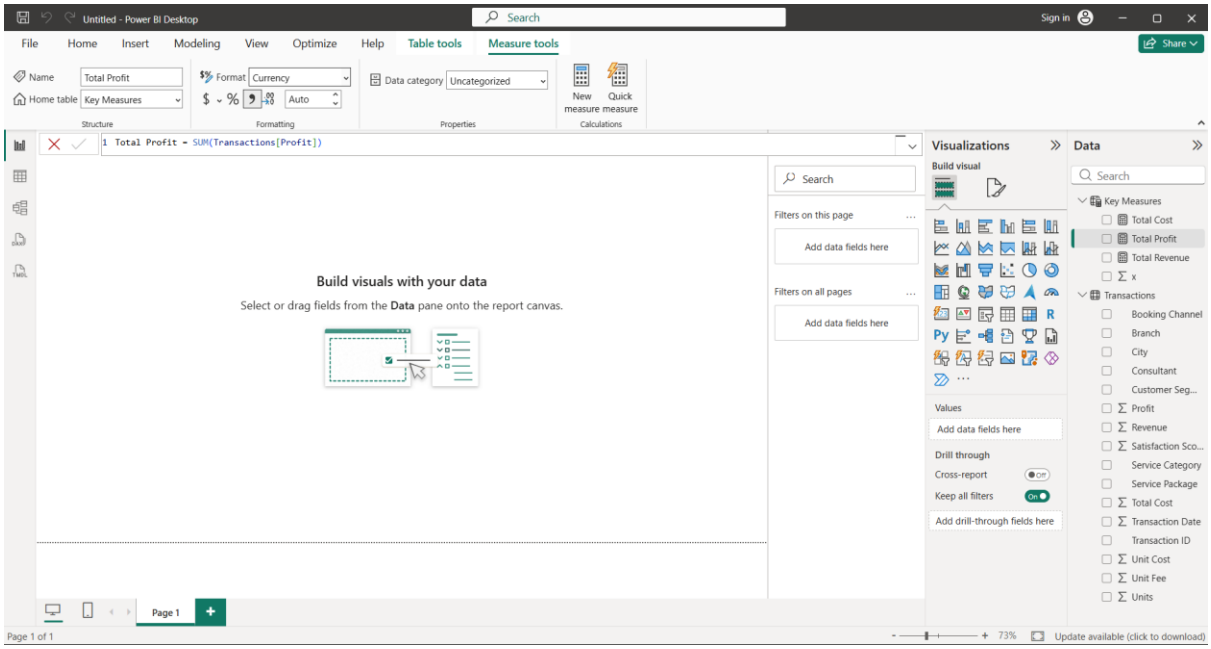
1. Total Revenue



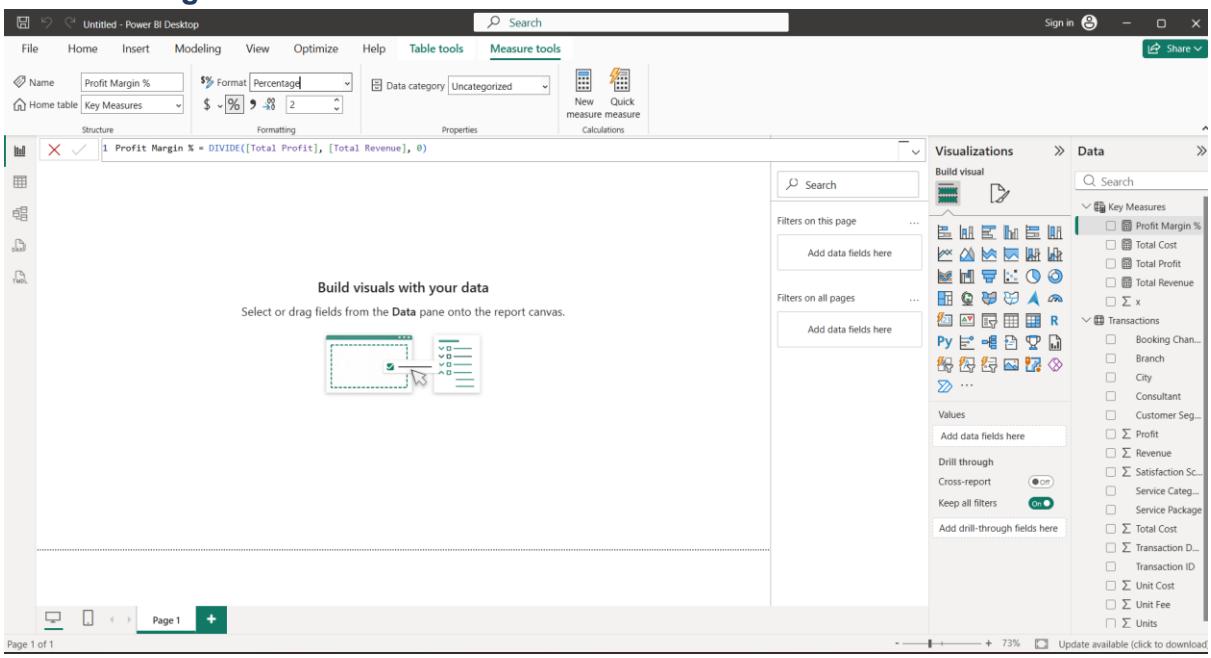
2. Total Cost



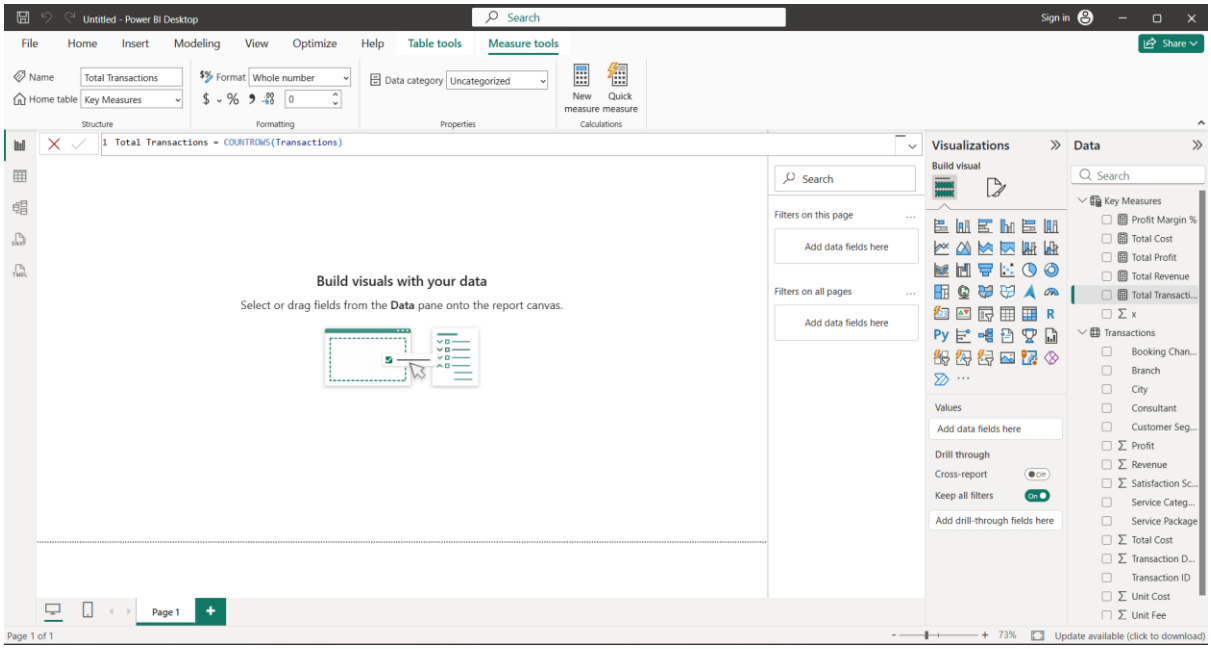
3. Total Profit



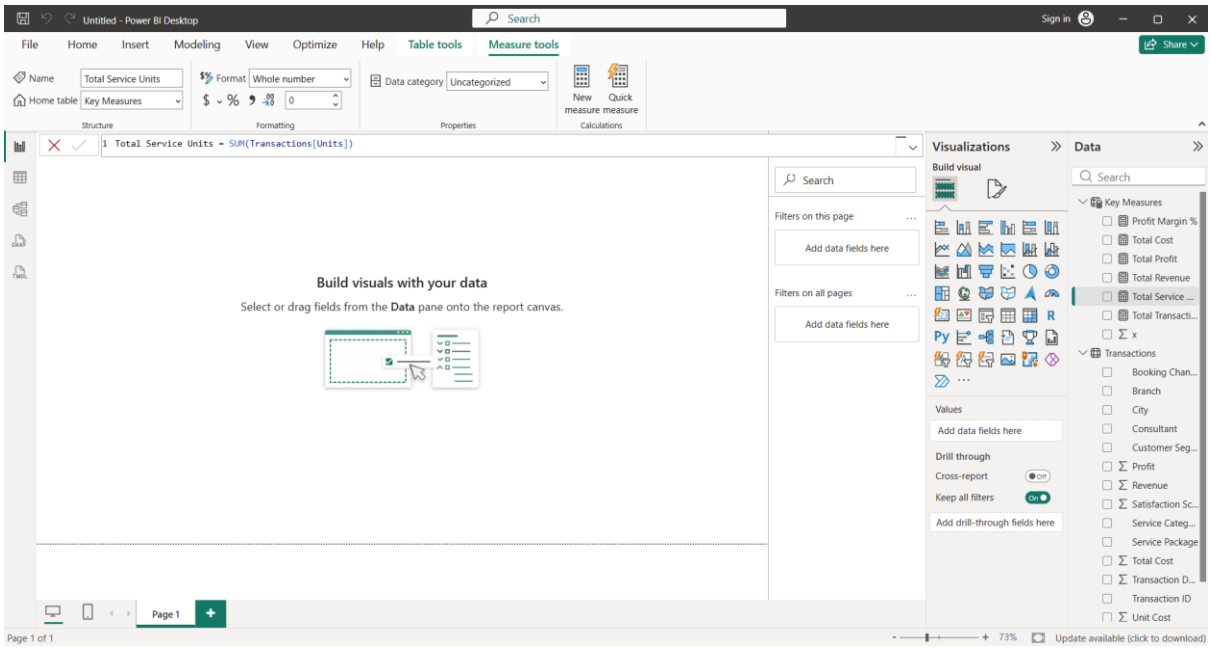
4. Profit Margin %



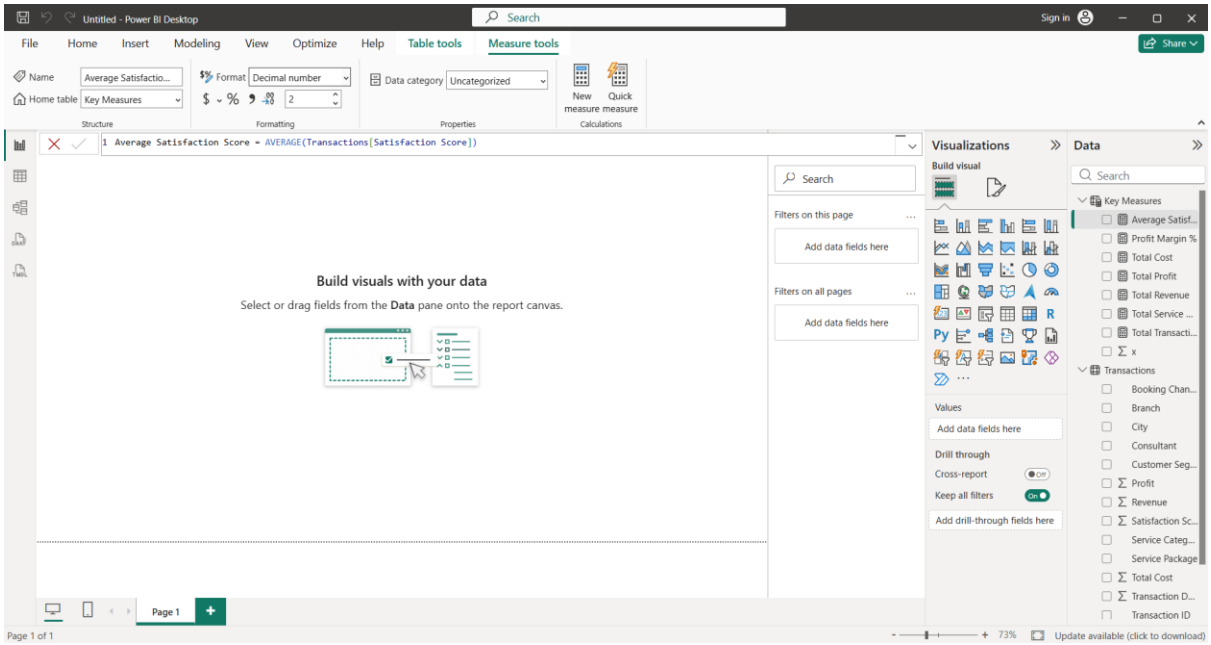
5. Total Transactions



6. Total Service Units

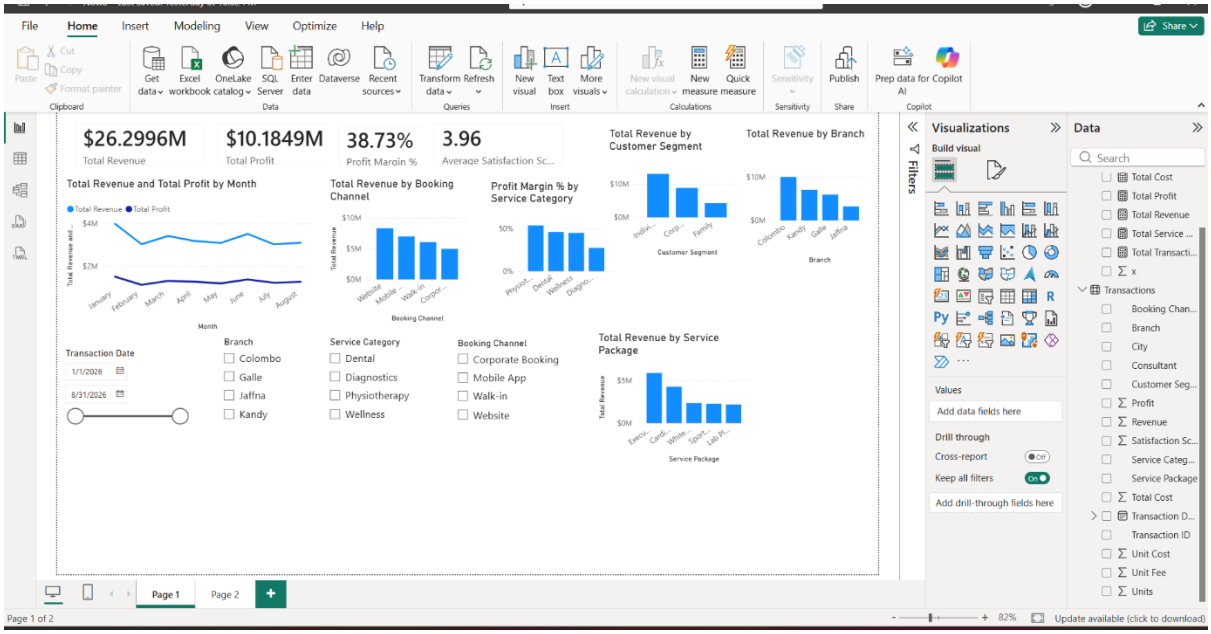


7. Average Satisfaction Score



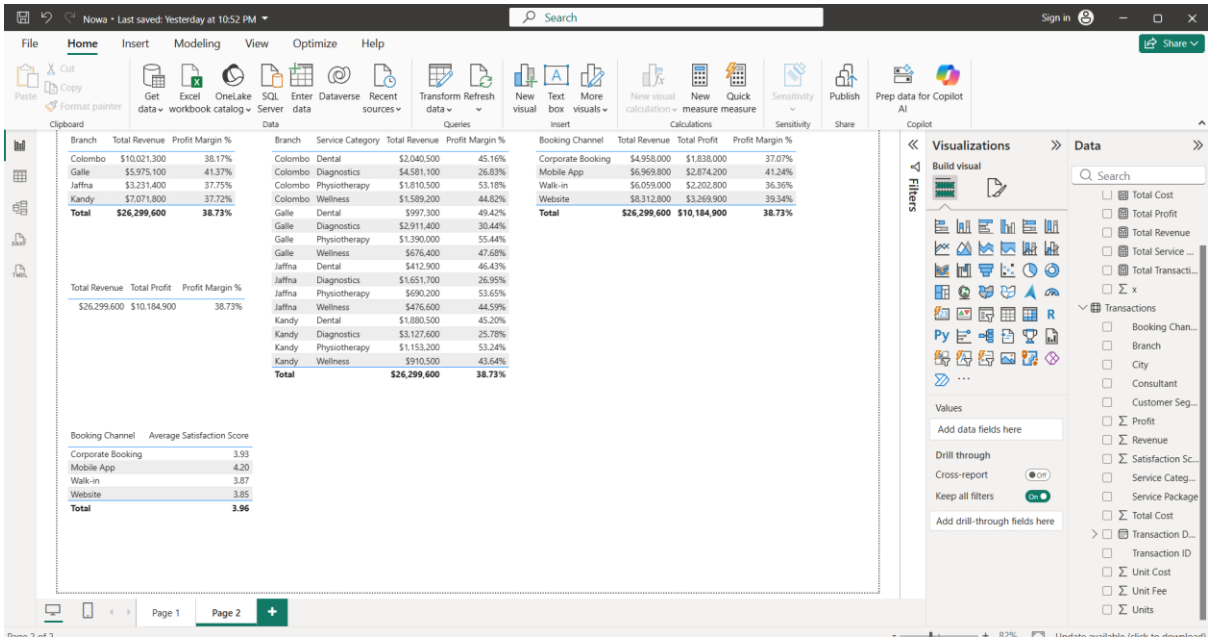
Interactive Dashboard

The final dashboard page brings together four KPI cards, six required charts, and four slicers (Date, Branch, Service Category, Booking Channel) on a single page for interactive exploration.



Part D — Business Analysis

Findings below are read directly from the dashboard visuals, plus three supporting tables built on a separate analysis page (Page 2) to get exact figures for margin- and satisfaction-based questions: Branch × Total Revenue × Profit Margin %, Branch × Service Category × Total Revenue × Profit Margin %, and Booking Channel × Total Revenue × Total Profit × Profit Margin % × Average Satisfaction Score. Include a screenshot of that analysis page alongside this report as supporting evidence.



Q1. Colombo generates the highest revenue at $10,021,300 (38.1% of the company total) — clearly ahead of Kandy ($7,071,800), Galle ($5,975,100), and Jaffna ($3,231,400).

Evidence: Branch × Total Revenue table: Colombo, $10,021,300; Kandy, $7,071,800; Galle, $5,975,100; Jaffna, $3,231,400; Company total, $26,299,600.

Q2. No. Colombo has the highest revenue but only a 38.17% profit margin — actually below the company average of 38.73%. Galle has the strongest margin at 41.37%, despite having the second-lowest revenue.

Evidence: Branch × Total Revenue × Profit Margin % table: Colombo 38.17%, Galle 41.37%, Jaffna 37.75%, Kandy 37.72%.

Q3. Diagnostics contributes the most revenue by a wide margin — $12,271,800 in total, or 46.7% of the company's $26.3M revenue, roughly double the next-highest category (Dental, $5,331,200).

Evidence: Branch × Service Category table, summed across all four branches: Diagnostics $12,271,800; Dental $5,331,200; Physiotherapy $5,043,900; Wellness $3,652,700.

Q4. Diagnostics — it is the single largest revenue contributor (46.7% of total revenue, Q3) but its blended profit margin is only about 27.4%, far below Physiotherapy's ~53.9%, Dental's ~46.1%, and Wellness's ~45.0%.

Evidence: Profit margins calculated from the Branch × Service Category table: Diagnostics generated roughly $3,367,000 profit on $12,271,800 revenue (~27.4%), the lowest ratio of any category.

Q5. Mobile App produces the strongest profitability, with a 41.24% profit margin — the highest of the four channels — even though Website generates more total revenue ($8,312,800 vs Mobile App's $6,969,000).

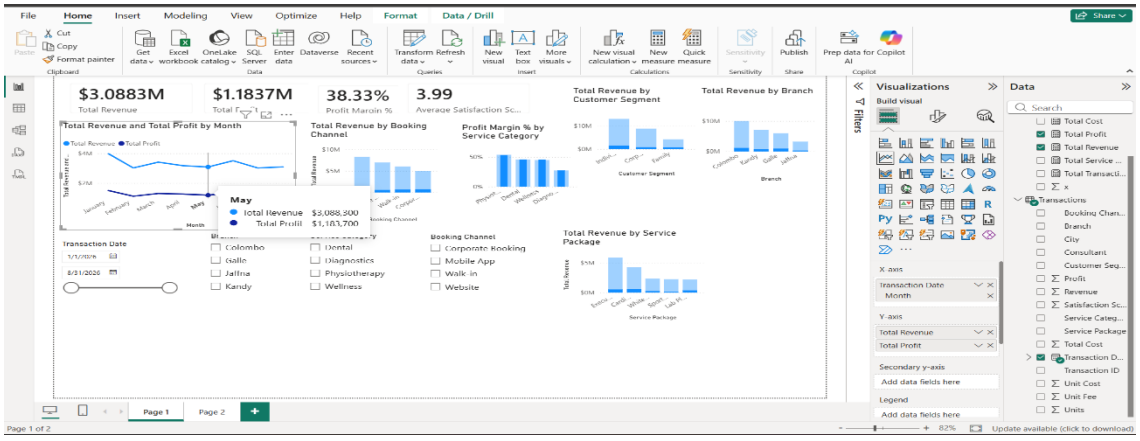
Evidence: Booking Channel table: Mobile App $6,969,000 revenue / $2,874,200 profit / 41.24% margin; Website $8,312,800 / $3,269,900 / 39.34%; Corporate Booking 37.07%; Walk-in 36.36% (lowest).

Q6. App again leads, with an average satisfaction score of 4.20 — notably above the company average of 3.96, and ahead of Corporate Booking (3.93), Walk-in (3.87), and Website (3.85, the lowest).

Evidence: Booking Channel × Average Satisfaction Score table: Mobile App 4.20; Corporate Booking 3.93; Walk-in 3.87; Website 3.85.

Q7. May 2026 shows the strongest overall performance. Clicking the May point on the trend chart cross-filters the page to show May's exact figures: Total Revenue of $3,088,300 and Total Profit of $1,183,700 — both noticeably above the other months on the same line chart — while the Profit Margin (38.33%) stays close to the company average (38.73%) rather than dropping, and the Average Satisfaction Score (3.99) is actually slightly above the overall average (3.96). So May wasn't just a high-revenue month with a discounted, low-margin mix — profitability and customer satisfaction both held up too.

Evidence: "Total Revenue and Total Profit by Month" line chart (Jan–Aug 2026), with the May data point selected: Total Revenue $3,088,300; Total Profit $1,183,700; Profit Margin % 38.33%; Average Satisfaction Score 3.99 (read from the KPI cards after cross-filtering to May).



Q8. Executive Screening, Cardiac Screening, and Whitening — the top three bars on the Top 5 Service Packages chart. Executive Screening and Cardiac Screening (both Diagnostics) account for a disproportionate share of total revenue, and Whitening is the strongest non-Diagnostics package, showing meaningful demand exists outside Diagnostics too.

Evidence: "Top 5 Service Packages by Revenue" chart, ranked left to right: Executive Screening > Cardiac Screening > Whitening > Sports Recovery > Lab Plus.

Q9. Kandy's Diagnostics line: it has the lowest profit margin of any branch/category combination in the whole business at 25.78%, on a substantial $3,127,600 of revenue — the second-highest Diagnostics revenue of any branch. High volume combined with the worst margin in the dataset makes this the clearest single combination to investigate.

Evidence: Branch × Service Category table: Kandy–Diagnostics 25.78% margin on $3,127,600 revenue, versus Colombo–Diagnostics 26.83%, Jaffna–Diagnostics 26.95%, and Galle–Diagnostics 30.44% — Kandy is the lowest of all 16 branch/category cells.

Q10. Diagnostics looks like the clear star performer on a revenue-only view — it generates 46.7% of all company revenue, nearly double the next category. But its actual profit margin (~27.4%) is the weakest of the four categories, roughly half of Physiotherapy's ~53.9%. Revenue-only reporting would encourage management to push even more Diagnostics volume, which would grow the top line while quietly dragging down overall company profitability.

Evidence: Cross-referencing the Branch × Service Category revenue totals (Q3: Diagnostics = $12,271,800, the largest category) against the derived category margins (Q4: Diagnostics ≈ 27.4%, the smallest) — revenue alone never surfaces this profitability gap.

Part E — Management Recommendations

Each recommendation follows the required structure: Finding → Evidence → Recommended Action, and is directly tied to dashboard visuals above.

Recommendation 1 — Fix the Diagnostics margin gap

Finding	Diagnostics generates 46.7% of total company revenue ($12,271,800), nearly double any other category, but its blended profit margin is only ~27.4% — well below Physiotherapy (~53.9%), Dental (~46.1%), and Wellness (~45.0%). Kandy's Diagnostics line is the single worst combination in the business at 25.78% margin on $3,127,600 of revenue.
Evidence	Branch × Service Category table: Diagnostics revenue and margin figures across all four branches, with Kandy–Diagnostics the lowest-margin cell (25.78%) in the entire 16-row matrix.
Recommended Action	Review the Diagnostics cost base branch by branch — starting with Kandy — covering consultant time allocation, equipment/lab costs, and pricing on Executive Screening and Cardiac Screening specifically, to lift margin without sacrificing the volume that makes Diagnostics the top revenue driver.

Recommendation 2 — Reduce revenue concentration in Colombo, and note Galle's leaner model

Finding	Colombo alone generates $10,021,300 of the company's $26,299,600 total revenue (38.1%), while Jaffna trails at only $3,231,400. Notably, Colombo's profit margin (38.17%) is actually below Galle's (41.37%), even though Galle earns far less revenue — Galle is running a more profitable model at smaller scale.

Evidence	Branch × Total Revenue × Profit Margin % table: Colombo $10,021,300 / 38.17%; Galle $5,975,100 / 41.37%; Jaffna $3,231,400 / 37.75%; Kandy $7,071,800 / 37.72%.
Recommended Actionn	Study what Galle is doing differently (service mix, staffing, cost control) that produces a higher margin on less revenue, and apply those practices to Colombo and Kandy. Separately, invest in growing Jaffna's revenue base to reduce single-branch dependency.

Recommendation 3 — Shift more bookings toward the Mobile App channel

Finding	Mobile App is the standout channel on both dimensions that matter: it has the highest profit margin of any booking channel (41.24%, versus 39.34% for Website, 37.07% for Corporate Booking, and 36.36% for Walk-in) and the highest average customer satisfaction (4.20, versus a 3.96 company average).
Evidence	Booking Channel table: Mobile App 41.24% margin / 4.20 satisfaction — both the highest in their respective columns — while Website generates more raw revenue ($8,312,800) but at a lower margin (39.34%) and lower satisfaction (3.85).
Recommended Action	Increase marketing spend and in-app promotions to shift volume from lower-margin channels (especially Walk-in) toward the Mobile App, since it simultaneously improves profitability and the customer experience rather than trading one off against the other.